

Homological Approaches to Hybrid Dynamics I

William Kalies¹ *

Department of Mathematical Sciences
University of Toledo, Toledo, OH

Abstract

In this talk I will discuss current and future work to develop theoretical and computational approaches to obtain topological characterizations of dynamics for hybrid systems. First, we extend theoretical results from last year on the Conley index for hybrid systems that satisfy the trapping guard condition to develop a computational framework for combinatorial dynamics computations by discretizations in hybrid space. Second, we report on work using a convex optimization framework for identifying switching systems from trajectory data developed by Iwasaki et.al. to compute combinatorial dynamics for switching systems. Finally, I will discuss ongoing work to extend the TDA-based cycling signatures framework to hybrid systems.

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